Developing Standards-Compliant Documentation for OptArrow with Generative AI Tools

1. OptArrow Overview

OptArrow is an optimization integration engine designed to serve as a bridge between different programming languages. It connects users to various solvers (such as GLPK, HiGHS, and Gurobi) regardless of the programming language used by the user. By leveraging Apache Arrow for zero-copy data exchange, it eliminates cross-language communication bottlenecks.

2. Project Objective: Audit-Readiness

The primary goal of this project is to produce **audit-ready process documentation** for OptArrow. This project focuses specifically on the **ISO/IEC TS 33061:2021** standard.

Each process within the standard requires specific evidence to meet formal requirements and achieve defined outcomes. To document OptArrow effectively, we must answer the **Base Practice (BP) questions** using evidence identified within the OptArrow resource hub (e.g., the GitHub repository).

The AutoDoc Opportunity

The current repository is missing several administrative and process-oriented artifacts required for a full audit. **AutoDoc**, an automated documentation service, can be enhanced to close these gaps. Currently, AutoDoc automates repository analysis and docstring generation; this project provides the blueprint for it to generate **ISO-compliant regulatory documentation**.

3. Evidence Mapping & Gap Analysis

Process: System/Software Requirements Definition Process (Section 5.5.4)

BP1: Elicit stakeholder requirements

- **Standard Activity:** Identify the individuals or groups who have a legitimate interest in the system and elicit their needs, expectations, and constraints.
- **Audit Logic & Explanation:** This practice requires project-produced evidence that all stakeholders (e.g., users, developers, maintenance teams) have been identified and their requirements gathered. It also requires a **feedback loop** to ensure requirements were reviewed and approved by stakeholders for correctness.
- **Evidences Found:**
    - `Architecture.md`: Clearly states the "Purpose" of OptArrow, characterizing business needs and mission objectives.
    - `Software_Requirements_Specification_OptArrow.md`: Section 1 (System Overview) provide context for user needs.
- **Missing Evidences & Recommendations:**

    - **Missing:** A formal **Stakeholder Register** (List of roles/names and their interests).
    - **Missing: Elicitation Logs** and a **Feedback/Resolution Log** documenting stakeholder approval.
    - **Recommendation:** AutoDoc should generate `elicitation_and_feedback.md` for example.

BP2: Define system/software requirements

- **Standard Activity:** Transform the stakeholder requirements into a set of system/software requirements.
- **Audit Logic & Explanation:** This involves documenting requirements in a technical format that is uniquely identified and includes both functional and non-functional aspects.
- **Evidences Found:**
    - `Software_Requirements_Specification_OptArrow.md`: Formally defines functional requirements (FR1-FR4) with unique IDs.
    - `Architecture.md`: Details technical implementation requirements for Arrow IPC.
- **Missing Evidences & Recommendations:**
    - **Missing:** Explicit **Non-Functional Requirements (NFRs)** for Performance, Reliability, and Scalability.
    - **Missing:** A defined **Security Policy**.
    - **Recommendation:** AutoDoc should update the SRS or generate a `security.md` for example.

BP3: Analyze requirements

- **Standard Activity:** Analyze the software requirements for technical risk, feasibility, and impact on the operating environment.
- **Audit Logic & Explanation:** This ensures requirements are feasible and verifiable. It involves evaluating how requirements interact with the target environment (hardware, OS, etc.).
- **Evidences Found:**
    - `Architecture.md`: Evaluates technical feasibility of iterative session modes.
    - `testScript.py`: Serves as a prototype proof-of-concept for the Arrow gateway.
- **Missing Evidences & Recommendations:**
    - **Missing:** A **Formal Requirements Review Record**.
    - **Missing: Verification Criteria** defining exactly how each requirement will be tested.
    - **Recommendation:** AutoDoc should generate a **Requirements Analysis & Verification Matrix (RAVM)**.

BP4: Manage requirements

- **Standard Activity:** Manage changes to the requirements and ensure that the current status of requirements is known.
- **Audit Logic & Explanation:** This practice covers change control (versioning) and the **maintenance of bidirectional traceability** from stakeholder needs to code and tests.
- **Evidences Found:**
    - **Git Repository History**: Provides an audit trail of all changes made to the source files.
    - `Contributing.md`: Outlines general contribution rules.
- **Missing Evidences & Recommendations:**
    - **Missing:** A formal **Change Control Process** and a **Revision History Table**.
    - **Missing:** A formal **Requirements Traceability Matrix (RTM)**.
    - **Recommendation:** AutoDoc should generate `docs/iso_33061/traceability_and_change_management.md`.

4. Technical Specifications for AutoDoc Generation

To automate the closing of these audit gaps, AutoDoc should be configured to generate the following artifacts based on the repository state:

ISO BP	Target Artifact	Data Source for AutoDoc	Required Content for AI Generation
BP1	`elicitation_and_feedback.md`	Git commit authors, README, and project logs.	Stakeholder Register, Elicitation Summaries, Feedback Log.
BP2	`NFR_and_Security.md`	Architecture.md and library dependencies.	Measurable Performance Metrics, Security Posture.
BP3	`verification_matrix.md`	Mapping of `docs/*.md` to `tests/*.py`.	Requirement ID, Test File Path, Verification Method.
BP4	`traceability_matrix.md`	`git log` and directory structure analysis.	Bidirectional links from Needs

5. Conclusion

While OptArrow has strong technical documentation, it lacks the administrative evidence required by ISO 33061. By using the mapping above as a template, AutoDoc can be updated to automatically bridge these gaps, transforming OptArrow into a fully audit-ready software system.